

Lemma

Tags: Probability

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Bernoulli Factories

This Lemma gives a short introduction to *Bernoulli factories*: randomized algorithms that simulate a Bernoulli random variable with parameter $f(p)$ using only samples from a Bernoulli distribution with parameter p .

A Fair Coin. First, let's see a simple algorithm for simulating a Bernoulli sample with parameter p , given access to a Bernoulli distribution with parameter p .

Algorithm 1 Fair coin using unbiased coin (von Neumann trick)

Require: Sample access to a Bernoulli distribution B_p with parameter $p \in (0, 1)$

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1: while true do
2:   Draw independent samples  $X_1, X_2 \sim B_p$ 
3:   if  $(X_1, X_2) = (0, 1)$  then
4:     return 0
5:   else if  $(X_1, X_2) = (1, 0)$  then
6:     return 1
7:   end if
8: end while
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Theorem 1. Given sample access to a Bernoulli distribution with parameter p , Algorithm 1 returns a Bernoulli sample with parameter $1/2$. Moreover, it draws

$$\frac{1}{p(1-p)}$$

samples in expectation.

Proof. In a single repetition,

$$\begin{aligned}\Pr[\text{output } 0] &= (1-p)p = p(1-p) \\ \Pr[\text{output } 1] &= p(1-p)\end{aligned}$$

Therefore,

$$\Pr[\text{algorithm outputs}] = 2p(1-p).$$

Conditioned on the event that the algorithm outputs, we obtain

$$\Pr[\text{output } 1 \mid \text{algorithm outputs}] = \frac{p(1-p)}{2p(1-p)} = \frac{1}{2}.$$

Similarly,

$$\Pr[\text{output } 0 \mid \text{algorithm outputs}] = \frac{1}{2}.$$

Thus, the algorithm returns a Bernoulli sample with parameter $1/2$. Now, each repetition succeeds with probability

$$2p(1-p).$$

Hence, the number of repetitions until termination follows a geometric distribution with expectation

$$\frac{1}{2p(1-p)}.$$

Since each repetition uses exactly 2 samples from B_p , the expected number of samples drawn is

$$2 \cdot \frac{1}{2p(1-p)} = \frac{1}{p(1-p)}.$$

□

Bernoulli Factories. A *Bernoulli Factory* is a randomized algorithm that, given sample access to a Bernoulli distribution with parameter p returns a Bernoulli random variable with parameter $f(p)$ for some function f . The above algorithm implements constant function

$$f(p) = \frac{1}{2}$$

A central question in Bernoulli factories is determining which functions f can be simulated using only samples from B_p .

Bernstein polynomial. It is easy to see that using exactly n samples, one can simulate any function of the form

$$f(p) = \sum_{k=0}^n a_k \binom{n}{k} p^k (1-p)^{n-k},$$

where $a_k \in \{0, 1\}$ for all k (since probability of k successes equals $\binom{n}{k} p^k (1-p)^{n-k}$). Equivalently, f is a linear combination of the degree- n Bernstein basis polynomials

$$\binom{n}{k} p^k (1-p)^{n-k}.$$

Alternatively, one may write

$$f(p) = \sum_{P \in \mathcal{P}} a_P P,$$

where $\mathcal{P}$ denotes the set of degree- n Bernstein polynomials and $a_P \in \{0, 1\}$.

The above observation suggests that Bernoulli factories are inherently connected to polynomial structure. Indeed, probabilities generated using finitely many samples are combinations of terms of the form $p^k (1-p)^{(n-k)}$. Surprisingly, the following theorem shows that this viewpoint characterizes all Bernoulli factories.

Theorem 2 (Keane and O'Brien (1994)). *A function $f : S \subseteq [0, 1] \rightarrow [0, 1]$ admits a Bernoulli factory if and only if one of the following holds:
 f is constant, or
 f is continuous and polynomially bounded on S .*

A function f is polynomially bounded if there exists $n \in \mathbb{N}$ such that $f(p), 1 - f(p) \geq \min(p^n, (1 - p)^n)$ for all $p \in S$.